Ali Al-Kelabi

AI & Robotics | System Design | Software & Automation

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LinkedIn | GitHub | Portfolio

Professional Profile

Mechatronics Engineer and Master's student in Artificial Intelligence for Smart Sensors and Actuators with hands-on experience in **machine learning, computer vision, robotics simulation, sensor systems, industrial automation and software development**. I build Python-based applications and data pipelines, implement computer-vision models (YOLO, OpenCV), integrate cameras and sensors, work with ROS 2 and Gazebo, and develop automation and monitoring solutions. Research background in pulsed eddy current measurements, sensor/probe development and non-destructive testing, including a **co-authored peer-reviewed publication**. My profile combines engineering fundamentals with software and AI development, across academic research, industry-related projects, university projects, personal projects and voluntary engineering initiatives.

Education

Master of Engineering – Artificial Intelligence for Smart Sensors and Actuators **03/2025 – Present**
Technische Hochschule Deggendorf, Germany

- Current German grade average: **2.4** (interim average, not a final degree grade).
- All coursework and graduation requirements completed except the Master's thesis.
- Focus: advanced programming, artificial intelligence, edge devices, machine learning and system design, autonomous systems, model-based function engineering, deep learning, computer vision, network communication, big data and advanced sensors.

Bachelor of Engineering – Mechatronics **02/2019 – 02/2023**
IIUM University Malaysia

- Graduated **with distinction** (German-equivalent grade: **2.0**).
- Focus: intelligent control, engineering design, advanced mathematics, deep learning, computer vision, software engineering, statics, dynamics, PCB design, industrial automation, robotics, embedded systems and embedded programming, signal processing, control systems, electrical systems, and hands-on electronics and software development.

Professional Experience

Working Student – UiPath and C# Automation **06/2025 – 11/2025**
Technische Hochschule Deggendorf Germany

- Developed automation use cases and practical examples with UiPath and C#.
- Built web-scraping workflows and automated data-collection processes; designed data-processing workflows to collect, structure and manage information.
- Identified opportunities to reduce repetitive manual tasks through workflow automation.
- Supported the development of teaching and practical materials on automation.

Project Collaboration – Machine Learning and Computer Vision **03/2024 – 07/2024**
ICON Malaysia Malaysia

- Contributed to a camera-based violation-detection project using machine learning and computer vision.
- Worked with YOLO-based object detection to identify relevant objects and events in camera images; combined object detection with OCR-related approaches to extract information from visual data.
- Supported image-data preparation, annotation and model training in Python; explored visible characteristics and object-size information to support detection and classification.
- Stored and organised detection results in a database; took part in developing and evaluating the computer-vision workflow.

German Language Lecturer **10/2023 – 07/2024**
4C Jordan Jordan

- Taught German at A1, A2 and B1 levels: grammar, vocabulary, pronunciation and sentence structure.
- Prepared learning activities for different proficiency levels and developed students' reading, writing, listening and speaking skills, adapting exercises to individual progress.

Student Research Assistant – Sensor Systems and Non-Destructive Testing **11/2019 – 02/2023**
IIUM / Malaysian Nuclear Agency Malaysia

- Conducted research on **pulsed eddy current (PEC)** measurements for non-destructive testing with experimental sensor systems.
- Used **LabVIEW** for measurement acquisition and experimental data collection.
- Contributed to the development, adaptation and testing of measurement probes.
- Supported experimental preparation, measurement procedures, data organisation and data preparation for analysis.
- Contributed to research resulting in a co-authored publication on machine-learning-based evaluation of corrosion under insulation in ferromagnetic structures (see Publication).

Engineering Intern – Mechanical and Energy-Related Investigation **07/2022 – 09/2022 (8 weeks)**
SURECO SDN. BHD. Malaysia

- Took part in an engineering investigation on vortex generators and engine configuration, aimed at reducing fuel consumption and improving energy output.
- **Led a student team** during the internship project; supported planning, coordination of project activities and equipment requirements.
- Contributed to technical requirements, engineering documentation and technical reports; evaluated possible engineering improvements with the team.

Web Development Project – Training-Based Collaboration **11/2020 – 12/2020**
Laimoon-associated project team Remote

- Developed a graphical registration form and web interface, and worked on collecting, parsing and managing submitted data.
- Used HTML, CSS, JavaScript, Node.js and Python with REST APIs, JSON and PostgreSQL; connected web interfaces to backend data processing.

Training-based project collaboration, not paid employment at Laimoon.

Research and Academic Projects

Gesture-Controlled UR5e Pick-and-Place System *Personal project – robotics simulation*
Tech: ROS 2, Gazebo, MoveIt 2, UR5e, Python, OpenCV, MediaPipe, YOLOv8, RGB-D camera, Docker

- Built a simulated robotic pick-and-place system controlled via hand gestures and visual interaction; integrated a simulated UR5e arm with ROS 2, Gazebo and MoveIt 2.
- Implemented gesture recognition (MediaPipe, OpenCV) and HSV-based object detection; used RGB-D data to estimate depth and convert image coordinates into 3D world positions.
- Developed a browser-based click-to-point dashboard; integrated collision-checked inverse kinematics and motion planning; integrated YOLOv8 detections via ROS 2 messages.
- Reported maximum object-position error of approx. **2 mm (x/y) and 1 mm (z)** within the simulation. **Repository**

Developed and evaluated in simulation; not a physical robot deployment.

YOLOv8 and ROS 2/Gazebo Integration
Tech: Python, YOLOv8, ROS 2, Gazebo, Computer Vision

- Connected YOLOv8 object detection with ROS 2 and Gazebo: camera images as inference input, detection outputs published as structured ROS 2 messages.
- Explored how computer-vision results can be used inside a larger robotics software architecture.

Agentic Workflow for Robotic Task Control *In development*
Tech: LangGraph, n8n, ROS 2, Gazebo, LLMs, Computer Vision

- Exploring an agentic workflow that translates natural-language instructions into structured robotic tasks, combining LLM-based task planning with ROS 2 and Gazebo.
- Investigating multi-step orchestration with LangGraph and n8n, and computer vision for object perception and task verification (instruction → task decomposition → robot action → visual feedback).

In development; not yet a completed end-to-end autonomous system or a vision-language-action model.

ROS-Based UR5 Robotic Manipulator Simulation
Tech: ROS 1, MoveIt, Gazebo, Python, C++, Ubuntu, HSV colour recognition

- Configured a simulated UR5 manipulation and motion-planning workflow with ROS 1, MoveIt and Gazebo; implemented HSV-based colour recognition for visual object identification in pick-and-place tasks. **Repository**

Industrial Telemetry Anomaly Detection Workflow
Tech: Python, PySpark, SQL, Z-score, Isolation Forest, Docker, Linux

- Built a data-processing workflow for industrial telemetry, organised into reusable processing stages, with Docker/Linux development environment.
- Implemented and compared statistical (Z-score) and machine-learning (Isolation Forest) anomaly detection; explored alert generation and automation. **Repository**

Industrial Telemetry and Environmental Monitoring System
Tech: Python, Raspberry Pi, Ubuntu, SQL, LoRa, RFM95W 868 MHz, sensors

- Developed an IoT monitoring and control system on a Raspberry Pi with UV-intensity measurements and LoRa wireless communication (RFM95W, 868 MHz).
- Implemented rule-based control of blinds and LED lighting, plus a touchscreen GUI with graphs, system information, sensitivity settings and alerts.

Diabetic Retinopathy Detection Using Deep Learning
Tech: Python, TensorFlow, Keras, OpenCV, ResNet50, DenseNet121

- Developed image classification and grading of retinal images with OpenCV preprocessing and transfer learning (ResNet50, DenseNet121); trained, evaluated and compared models using accuracy, macro recall, F1-score and quadratic weighted kappa (QWK). **Repository**

Technical project; not a clinically validated diagnostic system.

AML Alert Triage Copilot
Tech: Python, FastAPI, Streamlit, LLMs, Pydantic, structured JSON

- Prototype for reviewing and triaging *synthetic* anti-money-laundering alerts: LLM-generated structured explanations and recommendations, validated with Pydantic, FastAPI backend and Streamlit UI.

- Human-in-the-loop design with traceable, structured outputs instead of fully autonomous decisions. **Repository**

Local Research Assistant – Retrieval-Augmented Generation
Tech: Python, PyTorch, Hugging Face Transformers, Chroma, FastAPI, Docker

- Local RAG prototype that answers questions from personal notes and document collections: document processing, retrieval with a Chroma vector database, and language-model responses; FastAPI interfaces and Docker environment. **Repository**

CampaignPulse – Marketing Analytics Dashboard
Tech: Python, FastAPI, Google Analytics 4, Streamlit, Plotly

- Analysed Google Analytics 4 event data (purchases, revenue, user activity) and built interactive Plotly visualisations of campaign adoption, engagement and conversion in a Streamlit interface with FastAPI. **Repository**

Aerial Wildfire Detection Using Deep Learning *University course project, team of four, 10/2025 – 01/2026*
Tech: Python, TensorFlow, Keras, MobileNetV2, VGG16, NumPy, Matplotlib, Scikit-learn, Streamlit

- Classified satellite images into wildfire / non-wildfire: preprocessing, MobileNetV2 transfer learning, data augmentation, fine-tuning, evaluation and performance analysis.
- Contributed to the technical report, presentation and a Streamlit demo interface.

Report stated accuracy above 98% as a course-project result, not evidence of operational wildfire-detection performance.

IoT-Based Hydroponic System *University team project, **team leader**, 10/2021 – 05/2022*
Tech: Arduino Mega/Uno, ESP32, pH, turbidity, humidity and water-level sensors

- Led a student team building an IoT monitoring and control system: pH, turbidity, humidity and water-level sensing, automated water and fertiliser management concepts, pH control, Wi-Fi remote monitoring and a status display.
- Took part in sensor calibration, system testing and debugging.

Attendance Recognition Application *University project, **team leader**, 10/2022 – 12/2022*
Tech: Python, PyTorch, NVIDIA Jetson Nano

- Led a team developing a computer-vision attendance application on a Jetson Nano that recognises individuals and updates attendance records in a database.

Intelligent Control CNN Traffic-Control Competition *IIUM competition, **team leader**, 12/2022*
Tech: MATLAB, CNN, Arduino

- Led a team applying a CNN approach to traffic control, developed in MATLAB and demonstrated with Arduino – **2nd place**.

Water Quality Measurement System with GPS *University project, 11/2019 – 02/2020*
Tech: Arduino, pH sensor, turbidity sensor, GPS

- Co-developed a portable water-quality measurement system: pH and turbidity sensors, calibration, and GPS tagging of measurements with their location.

Research Publication

Ali Fadhil Al-Kelabi et al., "Machine-Learning-Based Evaluation of Corrosion Under Insulation in Ferromagnetic Structures." *IIUM Engineering Journal*, Vol. 22, Issue 2, pp. 226–233, 4 July 2021. DOI: 10.31436/iiumej.v22i2.1692

- **My contribution:** PEC measurements, LabVIEW-based data acquisition, probe development and adaptation, and experimental data preparation.

Technical Skills

Programming & Software: Python, C, C++, C#, JavaScript, SQL, Bash, HTML, CSS, Node.js, Git, Docker, Linux/Ubuntu, VS Code, REST APIs, JSON, FastAPI, Pydantic

AI & Machine Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, YOLO/YOLOv8, classification, object detection, transfer learning, fine-tuning, data augmentation, ResNet50, DenseNet121, MobileNetV2, VGG16, LLM applications, RAG, prompt engineering, LangGraph, n8n

Computer Vision & Perception: OpenCV, RGB-D / depth data, Intel RealSense D435i, image preprocessing and segmentation, HSV-based detection, OCR, 3D position estimation, image-to-world mapping, MediaPipe, gesture recognition

Robotics & Automation: ROS 2, ROS 1, Gazebo, MoveIt 2, UR5e simulation, manipulation and pick-and-place, inverse kinematics, collision-checked motion planning, UiPath, C# automation, workflow automation and process optimisation

Sensors, Embedded & Electronics: Arduino Uno/Mega, ESP32, Raspberry Pi, NVIDIA Jetson Nano, LoRa/RFM95W 868 MHz, sensor integration and calibration, LabVIEW, pulsed eddy current measurement, probe development, PCB design, edge computing, telemetry

Data Engineering & Analytics: Pandas, NumPy, PySpark, PostgreSQL, Microsoft SQL Server, SQLite, data pipelines, time-series data, anomaly detection (Z-score, Isolation Forest), Grafana, KPI development, dashboards, Plotly, Streamlit

Web & Application Development: React, Vue, Streamlit, FastAPI, Node.js, REST API development, structured JSON processing, database integration

Leadership and Voluntary Experience

Co-Founder – Arobot Malaysia **08/2021 – 08/2024**
Voluntary educational initiative in robotics and mechatronics Malaysia

- Co-founded an initiative for robotics, mechatronics and practical engineering learning; reached approx. **150 unique participants**.
- Organised and supported seminars, courses and workshops; supported Arduino and Raspberry Pi projects; contributed to starter kits and educational materials, illustrations and technical review.
- Contributed social-media content and video editing; supported communication with participants.

Voluntary initiative, not a registered-company employment position.

Robocon University Team (voluntary member) **04/2021 – 02/2023**
International Islamic University Malaysia (IIUM) Malaysia

- Worked on electronics, coding and testing of an arrow-shooting robot with mecanum-wheel mobility; supported development and testing of robot subsystems and hardware–software integration.

Graphic Design and Photography Volunteer **02/2021 – 02/2022**
IIUM Student Association Malaysia

- Created event photography, graphic design and communication content; documented university events, including a visit by the German ambassador.

Awards and Achievements

- **2nd Place – IIUM Intelligent Control CNN Traffic-Control Competition**, 12/2022 (team leader)
- **2nd Place – EDUTECH Competition**, 2021 (team member, water-surface-cleaning boat project)
- **IIUM Dean's List**, twice during the first two semesters of the Bachelor's degree, 2019–2020
- **IIUM Ummatic Scholarship**, 2020–2022 (full scholarship covering two years)
- Arrow Throwing Robot Competition, IIUM, 09/2021 (certificate of participation)

Certifications

- IBM – Python for Data Science and AI, 07/2024
- IIUM Professional Certificate – Deep Learning in Video Analytics and Computer Vision, 12/2022
- NVIDIA – Getting Started with AI on Jetson Nano (Certificate of Competency), 12/2022
- Laimoon – Ultimate Guide to Serverless JavaScript Development on AWS, 11/2020 (Credential ID: LVC-FJ94DDT35)

Languages

English: C2 (fluent) • **German:** B2–C1 (fluent)

13.11.2026

